

Unifying Speech and Language Through Continuous Dual-Fusion for Medium-Resource Languages: The Cases of Hungarian and Greek

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Abstract—In medium-resource languages like Hungarian and Greek, Automatic Speech Recognition (ASR) suffers from scarce paired acoustic data, whereas abundant text data can be leveraged via Large Language Model (LLM) priors. To compensate for this acoustic deficit, we propose an advanced Dual-Fusion architecture that shifts ASR from isolated acoustic decoding to operating entirely within an intermediate audio-conditioned text space. Our model bridges a frozen acoustic encoder (Whisper) with pre-trained LLMs via both prompt-level embeddings and deep, continuous-space alignment across multiple intermediate decoder layers. However, unconstrained LLM priors often over-power limited acoustic signals, causing the model to over-correct grammar and drop dysfluencies. To actively control this semantic-acoustic trade-off, our methodology introduces asymmetric down-sampling to preserve fine phonetic boundaries, multi-layer gated cross-attention, target text perturbation, and a strict two-stage LoRA curriculum. Evaluated on diverse Hungarian and Greek datasets, our framework significantly reduces Word Error Rate (WER) compared to standalone decoders to both standalone acoustic decoders and acoustic-linguistic models like Vox-KriKri, demonstrating generalized performance improvements across both languages.

Index Terms—Automatic Speech Recognition, Speech LLMs, Continuous Fusion, Medium-Resource ASR, Multimodal Alignment

I. INTRODUCTION

Automatic Speech Recognition (ASR) has evolved significantly with the advent of robust end-to-end models like Whisper [1] and highly data-efficient architectures like Canary [2], and massively multilingual systems like MMS [3], which achieve state-of-the-art multilingual performance. However, empirical evaluations reveal that these systems frequently exhibit performance degradation when applied to medium-resource and morphologically complex languages, such as Hungarian and Greek. This drop in accuracy is rarely due to a failure in acoustic feature extraction; rather, it manifests as orthographic and morphological errors stemming from a lack of deep, language-specific linguistic priors [4].

To bridge this gap, recent research has shifted toward multimodal Speech-Language Models (SLMs) that couple pre-trained acoustic encoders with Large Language Models (LLMs) [5]. Broadly, this literature has bifurcated into two families of integration strategies [6]: prompt-level (GPT-style) approaches that prepend compressed acoustic embeddings as pseudo-text tokens, and cross-attention (T5-style) approaches

that let text states attend to acoustic representations at intermediate layers. Early prompt-level strategies sought to minimize architectural complexity: Fathullah et al. [7] showed that prepending compressed conformer embeddings to a frozen LLaMA backbone already yields competitive ASR, while the SLAM-ASR framework [8] demonstrated that a simple, trainable linear projector connecting an encoder like Whisper to an LLM could yield highly competitive ASR capabilities. Expanding toward broader auditory comprehension, models like SALMONN [9] utilized Q-Formers to compress and bridge audio embeddings into the LLM space. However, rigid temporal pooling risks destroying the micro-phonetic boundaries required for highly inflected languages [10].

To achieve deeper semantic integration, researchers explored cross-attention fusion. The Whispering LLaMA framework [11] pioneered cross-modal generative error correction by integrating Whisper’s audio embeddings directly into the LLM via cross-attention adapters. Comparative studies on connector structures confirm that while multi-head cross-attention introduces more parameter overhead than simple linear layers, it provides significantly superior representational alignment [12]. Furthermore, foundational vision-language research, such as the Flamingo [13], and the Whisper-Flamingo [14] models established the efficacy of gated cross-attention to interleave frozen text blocks with continuous multimodal inputs, ensuring stability during generation. In the speech domain, the recent VoxKriKri [15] framework operates fully within a continuous latent space, fusing Whisper’s hidden decoder states with an LLM’s textual representations at designated layers, proving that intermediate layer injection yields robust cross-modal alignment for languages like Greek.

A related question is which encoder representations to expose. Rather than passing only an encoder’s final hidden state, SUPERB [16] and follow-up work [17] establish that a learned weighted combination over all encoder layers outperforms single-layer extraction. More recent work replaces static weights with input-dependent attention: Adaptive Layer Attention [18] and TASLA [19] support layer-wise fusion via learned attention weights over fixed averaging, while LayAlign [20] shows a closely analogous attention-weighted layer-fusion mechanism improves multilingual reasoning in text-only LLMs – evidence that this design generalizes across modalities, although it has not previously been applied to

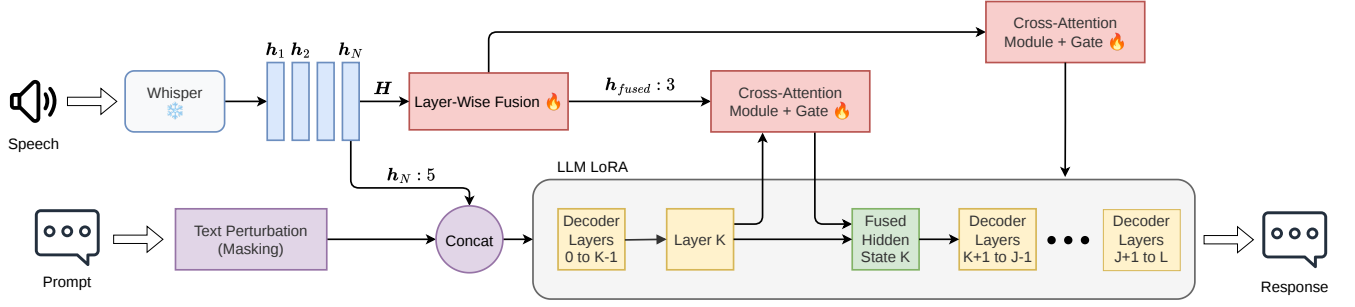


Fig. 1. Dual-fusion architecture: prompt injection, cross-attention, gated fusion, multiple injection layers, and layer-wise fusion.

acoustic-linguistic fusion for ASR.

A third design question concerns how to adapt the frozen LLM once bridged to the acoustic encoder. Naively fine-tuning LoRA-adapted LLM weights jointly with a freshly initialized bridge network risks the backbone compensating for a poorly aligned interface before either component stabilizes. WavLLM [21] addresses an analogous risk with an explicit two-stage curriculum – first building elementary cross-modal competence, then activating LoRA for complex multi-task instruction tuning – though only for high-resource, English-centric tasks.

Finally, a persistent failure mode in fused acoustic-LLM decoders is the “lazy LLM” problem, where the decoder ignores its acoustic information and defaults to its text prior. Text perturbation techniques mitigate this by deliberately weakening the model’s access to clean linguistic context during training [22]

Building on these findings, we propose a Dual-Fusion architecture that couples a frozen Whisper encoder with language-specialized LLM backbones for Hungarian and Greek. Our methodology combines: (i) a Dual Fusion Strategy feeding acoustic information at both the prompt level and via cross-attention with decoupled downsampling factors; (ii) Multiple Injection Layers with Layer-Wise Fusion, distributing cross-attention across intermediate decoder layers while aggregating all Whisper encoder layers through a dynamic, MLP-based attention mechanism instead of using only the final hidden state; (iii) Gated Cross-Attention, scaling injected context with a learnable tanh gate so the LLM can shield purely syntactic generation from acoustic noise; (iv) a 2-Stage LoRA Curriculum, first training only the bridge network (projectors, downsamplers, cross-attention modules) then activating LoRA to adapt the LLM network end-to-end; and (v) Text Perturbation, masking random tokens in the teacher-forced target sequence, directly counteracting language-prior dominance.

We evaluate this methodology separately on Hungarian (Common Voice, FLEURS, Speech Massive, VoxPopuli, Yodas, DataOcean, and Datatang) and Greek (Common Voice, FLEURS, HParl, TEDx, Logotypographia, and Stoma) corpora, using *elte-nlp/Racka-4B* [23] and *Llama-KriKri-8B* [24] as the respective linguistic backbones. Compared against stan-

dalone acoustic decoders and earlier acoustic-linguistic fusion baselines such as VOX-KriKri, our Dual-Fusion models yield consistent Word Error Rate reductions across both languages

II. METHODOLOGY

The methodology couples a frozen Whisper encoder with a language-specific LLMs (Racka-4B for Hungarian, Llama-KriKri-8B for Greek) through prompt-level conditioning and intermediate cross-attention, then adapts only lightweight modules with staged LoRA to preserve linguistic priors while improving low-resource ASR grounding. An overview is given in Fig. 1.

A. Dual Fusion Strategy

To effectively bridge the temporal mismatch between high-frequency acoustic frames and dense semantic tokens, we deploy a Dual-Fusion Strategy featuring asymmetric downsampling. Acoustic embeddings are integrated at two distinct points: the prompt level and the intermediate cross-attention layers. The first fusion path is prompt injection, where highly compressed acoustic representations (downsampling factor of 5) are projected through a Simple Reshaping adapter into the LLM input space as prefix-like speech tokens. The second fusion path is cross-attention injection, where text states attend to speech-derived keys and values at selected decoder layers. To prevent the destruction of micro-phonetic boundaries—such as brief Hungarian suffixes or rapid Greek vowel transitions—we deploy a learnable Conv1D downsampler with a smaller downsampling factor of 3, preserving fine-grained acoustic information for cross-attention.

B. Multiple Injection Layers as Persistent Acoustic Memory

VoxKriKri searches for a single best injection depth. We instead inject cross-attention simultaneously at several intermediate layers (e.g., [10, 20, 30] for KriKri, [15, 25, 35] for Racka), since a single early exposure risks being diluted as the hidden state propagates deeper. Repeated injection maintains a persistent acoustic memory, continually re-grounding generation instead of drifting toward pure language-model continuation.

C. Layer-Wise Fusion of Acoustic Representations

VoxKrikri [15] fuses only Whisper’s final hidden state; we instead stack all encoder layers $h_1, \dots, h_N$, since lower layers preserve phonetic detail and higher layers carry semantic abstraction — both valuable under scarce paired supervision. A *LayerWiseAttention* module first mean-pools each layer’s representation over the time dimension, then replaces a fixed average with input-dependent weights:

$$\mathbf{h}_{fused} = \sum_{l=1}^N \alpha_l \mathbf{h}_l, \quad \alpha_l = \text{softmax}(MLP(\mathbf{h}_1; \dots; \mathbf{h}_N))$$

D. Gated Cross-Attention

Injecting acoustic context at multiple layers along two pathways increases the risk that acoustic signal overwhelms the LLM’s own generative flow, especially for tokens that are purely syntactic (punctuation, function words) and should be governed by linguistic, not acoustic, evidence. Following the zero-initialized gating principle introduced for cross-modal residuals in Flamingo [13], each injection layer scales its cross-attention output with a learnable tanh gate before adding it to the hidden state:

$$\mathbf{h}_{fused} = \mathbf{h}_{LLM} + \tanh(\mathbf{g}) \odot \mathbf{C}_{audio}$$

E. Two-Stage LoRA Curriculum

Training new modules jointly with LoRA-adapted [25] LLM weights from initialization risks the backbone compensating for a poorly aligned bridge. We therefore freeze the LLM and Whisper in Stage 1, training only the projection, downsampling, and cross-attention modules; Once the bridge network establishes a stable interface, LoRA matrices are activated on the attention and MLP projections of the LLM decoder (query, key, value, output, and gated MLP up/down projections).

F. Text Perturbation

To counter the “lazy LM” problem, where teacher-forced text alone lets the decoder ignore acoustic cross-attention, we randomly replace a fraction p of target tokens with the tokenizer’s pad token in the teacher-forced decoder input (with $p = 0.1$ in our experiments), while leaving the true target labels used for the loss unperturbed. The corrupted context can no longer trivially predict the masked positions, forcing cross-attention to actively query the acoustic representations to reconstruct them.

III. EXPERIMENTAL SETUP

A. Backbones and Injection Configuration

Both language configurations adopt the frozen Whisper-large-v3 [1] encoder (636M parameters). For Greek, the LLM backbone is Llama-KriKri-8B [24] (8B); for Hungarian, eltenlp/Racka-4B [23] (~4.0B) (inherits the architecture directly from the Qwen/Qwen3-4B [26] model). The trainable bridge — reshape adapter (19M), a Conv1D downsampler (5M) shared across all injection layers (Section II-B), cross-attention

modules (59M), and layer-fusion module (1M) — totals 83M parameters (0.9% and 1.75% of the full model in Greek and Hungarian respectively), making Stage 1 substantially cheaper than full fine-tuning. Following Section II-B, cross-attention is injected at three intermediate decoder layers, chosen relative to each backbone’s depth: [10, 20, 30] for KriKri (32 layers) and [15, 25, 35] for Racka (36 layers). Each cross-attention module uses 32 attention heads. The prompt-level path compresses the final acoustic embedding $\mathbf{h}_N$ by a factor of $r_p = 5$, while the cross-attention path retains an effective compression factor of $r_c = 3$ for the fused acoustic embedding $\mathbf{h}_{fused}$.

B. 2-Stage LoRA Curriculum and Training Hyperparameters

Training follows the strict two-stage curriculum of Section II-E. In Stage 1, the LLM and Whisper encoder are fully frozen and only the bridge network is optimized, for 45000 steps, with learning rate 10^{-4} , an effective batch size of 64 (per-device batch size 4 with 4 gradient-accumulation steps across 4 GPUs), bf16 mixed precision, and the AdamW [27] optimizer under a linear scheduler with warmup. The LLM is fine-tuned under 4-bit NF4 quantization with double quantization (QLoRA-style [28]) to reduce memory footprint. In Stage 2, LoRA is activated on the LLM with rank $r = 16$, scaling $\alpha = 2r = 32$, dropout 0.1, targeting the attention projections and the gated-MLP up/down/gate projections [25]; the bridge network trained in Stage 1 (adapter, downsampler, and cross-attention modules) is frozen for the remainder of training so that LoRA adapts the LLM to the now-fixed acoustic interface rather than co-adapting with a still-moving bridge. This stage runs for 45000 steps, too, at learning rate 10^{-4} .

C. Corpora

Table III-D lists the Greek (Common Voice [29], FLEURS [30], HParl [31], TEDx [32], Logotypographia [33], Stoma [34]) and Hungarian (Common Voice [29], FLEURS [30], Speech Massive [35], [36], VoxPopuli [37], Yodas [38], DataOcean [39], Datatang [40]) corpora, spanning formal, read, and spontaneous registers. Each corpus is split into train/validation/test following a fixed priority: (1) the creators’ own official splits, if provided; (2) otherwise, a leave-subject-out (LSO) split by speaker/subject ID, if the corpus is subject-segmented, ensuring no speaker overlaps train and test, using a 60/20/20 ratio; (3) otherwise, a random split using a 80/20/20 ratio, seeded for reproducibility. Since dataset sizes vary by orders of magnitude, training batches are constructed via interleaved sampling across corpora with a temperature factor of 1.2, downweighting the largest datasets.

D. Preprocessing and Prompting

Transcripts are normalized before training/scoring: our full pipeline performs NFC unicode/symbol canonicalization, removal of speaker-tag prefixes, cutoff markers, transcriber annotations, and stray punctuation, Hungarian number verbalization, and hash-tag cleanup; Each utterance can additionally be conditioned on a natural-language epistemic

Dataset	Language	Duration (h)
Common Voice [29]	Greek, Hungarian	34h / 180h
FLEURS [30]	Greek, Hungarian	12h / 12h
HParl [31]	Greek	120h
TEDx [32]	Greek	30h
Logotypographia [33]	Greek	52h
Stoma [34]	Greek	23h
Speech Massive [35]	Hungarian	5.7h
VoxPopuli [37]	Hungarian	63h
Yodas [38]	Hungarian	150h
DataOcean ASR657/659 [39]	Hungarian	-
DataTang ASR1/2 [40]	Hungarian	-

prompt telling the model what transcription style to produce — [PROMPT_CLEAN] (accurate, clean transcript), [PROMPT_VERBATIM] (full, word-for-word transcript, disfluencies retained), or [PROMPT_FORMAL] (strictly formal/official transcript). Also, during training, the Whisper encoder’s input features are additionally regularized with SpecAugment [41], which we found helps prevent the bridge network from overfitting.

IV. RESULTS AND DISCUSSION

A. Greek: Comparison Against Baselines

Table I compares against full-context VoxKrikri [15] and Whisper-v3 (across 6 Greek Corpora). VoxKrikri is the first Greek speech LLM: it fuses Whisper large-v3’s decoder hidden states with Llama-KriKri-8B (a Greek-adapted Llama 3.1-8B) through a cross-modal attention module inserted at a single chosen decoder layer.

Our model wins decisively on Common Voice ($\Delta = -4.74$), HParl ($\Delta = -4.81$), and Logotypographia ($\Delta = -2.80$), but FLEURS is the one corpus where VoxKrikri’s simpler single-stage, single-injection design wins: on FLEURS, VoxKrikri (full) (9.33) outperforms ours (10.9, $\Delta = +1.57$), though we still beat Whisper-v3 (11.02) on this dataset.

TABLE I
GREEK WER (%) COMPARISON AGAINST BASELINES.

Dataset	Ours	Δ	Vox (full)	Whisper-large
Common Voice	7.3	-4.74	12.04	13.98
FLEURS	10.9	+1.57	9.33	11.02
HParl	8.2	-4.81	13.01	16.99
Logotypographia	6.6	-2.80	9.4	10.86
TEDx	17.8	-	-	-
GPC	-	-	12.08	17.27

B. Hungarian: Comparison Against Baselines

Table II reports and compares the WER scores across all nine Hungarian corpora against the original multilingual Whisper large-v3 [1]. We deliberately do not compare against the Hungarian-specific Whisper-Hu [42]; Whisper-Hu’s own model documentation discloses that it was fine-tuned using both training, test and validation splits of VoxPopuli, Common Voice, and Speech-Massive as training data, which makes any comparison against it on those benchmarks unfair rather

than zero-shot. For consistency, we restrict the comparison throughout to the original, unmodified Whisper checkpoint.

Our model wins outright on seven of the eight comparable corpora (Table II), with the largest margins on FLEURS (10.2 vs. 37.8, $\Delta = -27.6$) and Speech Massive (38.2 vs. 59.1, $\Delta = -20.9$), and further wins on Common Voice ($\Delta = -6.9$), VoxPopuli ($\Delta = -8.0$), DataOcean ASR 657 ($\Delta = -5.6$), DataOcean ASR 659 ($\Delta = -6.7$), and DataTang ASR2 ($\Delta = -2.8$). The sole loss is Yodas, where plain Whisper remains stronger (11.5 vs. 8.1, $\Delta = +3.4$). DataTang ASR1 (21.1) has no baseline comparison available.

TABLE II
HUNGARIAN WER (%) COMPARISON AGAINST THE ORIGINAL WHISPER BASELINE.

Dataset	Ours	Δ	Whisper-large
Common Voice	9.0	-6.9	15.9
FLEURS	10.2	-27.6	37.8
Speech Massive	38.2	-20.9	59.1
VoxPopuli	9.8	-8.0	17.8
Yodas	11.5	+3.4	8.1
DataOcean ASR 657	5.9	-5.6	11.5
DataOcean ASR 659	26.3	-6.7	33.0
DataTang ASR 2	6.8	-2.8	9.6
DataTang ASR 1	21.1	-	-

C. Ablation Study

D. Model Interpretability

E. Semantic Drift Analysis

F. Limitations

V. CONCLUSION

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